An approximate formula for the absorptivity of the three-body problem

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The absorptivity is the only piece of Kol's flux theory for the three-body problem that still has to be measured by simulation. Since it is fixed in the first passage, we decompose this passage into two rules: a mass rule, which decides between flying by and exchanging partners, and an energy rule, which decides whether the outgoing body stays bound to the pair. In 42,415 first passages in the plane with zero total angular momentum, the chance of staying bound falls with the binding of the pair and, in exchanges, with the mass gain of the new pair, according to logistic laws that explain up to 72% of the variation in mass combinations not used in the fit. The resulting formula, with eight constants fitted only on a mass grid, reproduces the measured absorptivity in seven channels outside this grid with R^2 of 0.72 to 0.77, on average over eccentricity, and, multiplied by the flux, predicts the escape chances with a mean error of 1.7 percentage points, against 4.2 for the volume theory and 2.5 for the flux without absorptivity. The dependence on eccentricity, strong in some channels, is captured only in part by the speed of the pair at apoapsis, and the distribution of the final eccentricity still requires the measured absorptivity.

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I. INTRODUCTION

In Kol's flux theory, the probability of each outcome of a three-body system is the product of a factor known in closed form, the phase-volume flux, and an absorptivity, which to this day must be measured by simulation [1, 3]. Note FR-TC-10 showed that the absorptivity of a whole run is fixed in the first passage of the incoming body through the pair: it is the chance that this passage ends in chaos or leaves the outgoing body bound [5]. Note FR-TC-11 showed that, in this passage, the choice between flyby and partner exchange follows a simple mass law [6]. To arrive at a formula for the absorptivity, the second half is missing: given who leaves, the chance that it stays bound to the pair instead of escaping. This note measures that chance, proposes approximate laws for it, including the effect of the eccentricity of the pair, and tests the resulting formula without simulating the cases to which it is applied.

II. METHOD

We use the 42,415 regular first passages (with at most one scramble) from the grids of note FR-TC-10 (eight mass channels) and note FR-TC-11 (28 mass combinations), in planar motion with zero total angular momentum, with three values of eccentricity (0.15, 0.5, and 0.85) and relative energies of the pair $x = E_B/E_0$ from 1.05 to 10 [5, 6]. The outgoing body stays bound when the energy of the pair after the passage, x_{novo}, is below the escape threshold, $x_{\text{novo}} < 1$. We separate the two branches: the flyby, in which the incoming body leaves, and the exchange, in which one of the binary members leaves. For each branch, we fit logistic models of the chance of staying bound and measure the predictive ability by leaving out, each time, one mass combination.

For the final test, we fit all the constants only with the mass grid of note FR-TC-11 and apply the formula to the seven channels of notes FR-TC-09 and FR-TC-10, whose mass combinations are not in that grid. We compare the absorptivity of the formula with the measured one in 924 energy and eccentricity cells [4, 5] and, multiplying it by the flux, we predict the escape chances and the distribution of the final eccentricity in the free falls of these sets.

III. RESULTS

A. What happens to the pair in each branch

Let $D = \ln(x_{\text{novo}}/x)$ be the logarithmic change in the energy of the pair in the first passage. In the flyby, the pair leaves on average less bound than it entered (median D of -0.14), and the effect grows with the binding of the pair. In the exchange, D depends mainly on the masses: if the new pair is heavier than the old one, it becomes more bound, and the released energy pushes the expelled body away. The simplest estimate, in which the new pair keeps the semimajor axis of the old one, gives $D_0 = \ln(m_1 m_2/m_a m_b)$, where m_1 and m_2 are the masses of the new pair; the measured median of D correlates with D_0 with $r = 0.77$. In the exchange between bodies of the same mass, the Newton's cradle effect of note FR-TC-11, $D_0 = 0$, and the pair hardly changes.

B. Laws for the chance of staying bound

Fitted only with the mass grid, the most compact laws are

$$P_{\text{bound}}^{\text{fly}} = \sigma\left(-0.24 - 1.62\ln x + 0.54\ln\frac{m_s}{m_a+m_b}\right), \qquad (1)$$

$$P^{\text{exch}}_{\text{bound}} = \sigma(-0.08 - 1.56\ln x - 1.06 D_0)\,, \qquad (2)$$

where $\sigma(z) = 1/(1+e^{-z})$ and m_s is the mass of the incoming body. Table I summarizes the predictive ability. The exchange law explains 72% of the variation among cells in a mass combination not used in the fit; the flyby law, 41% to 48%. The eccentricity hardly appears on average, but its effect is specific to each combination: in a heavy pair in a nearly circular orbit, a light body that flies by never stays bound, while, with the pair in an eccentric orbit, it sometimes does. The quantity that best captures this effect is the ratio between the flyby speed and the orbital speed of the pair at apoapsis, where the pair spends most of its time. With it, the flyby law rises to 54%, and the flyby or exchange law, evaluated in the same way, rises from 78% to 88%.

C. The complete formula

Combining the two parts, the absorptivity of an arrival with relative energy x is

$$\mathscr{A} = P_{\text{exch}}\,\bar{P}^{\text{exch}}_{\text{bound}} + (1 - P_{\text{exch}})\,P^{\text{fly}}_{\text{bound}}, \qquad (3)$$

$$P_{\text{exch}} = \frac{1}{1 + (0.52\, m_{\max}/m_s)^{1.9}}, \qquad (4)$$

with eight constants, all fitted on the mass grid. Here $\bar{P}^{\text{exch}}_{\text{bound}}$ is the mean of $P^{\text{exch}}_{\text{bound}}$ between the two possible cases, in which m_a leaves or m_b leaves, which expresses the coin toss between the binary members observed in note FR-TC-11, and $m_{\max}$ is the heavier member. The version with eccentricity uses the same three parts, with fifteen constants. The formula ignores direct chaos in the first passage, which ranges from 0% to 8% depending on the channel.

Applied to the seven channels that did not take part in the fit, the formula reproduces the measured absorptivity with $R^2 =$ 0.72 to 0.77 when the average over eccentricity is compared, and with $R^2 = 0.59$ and a mean error of 0.10 cell by cell (Fig. 1). The version with eccentricity does not improve the cell-by-cell comparison: the eccentricity effect learned on the grid does not transfer well to the other channels.

D. Escape chances without simulating the case

Multiplying the flux by the absorptivity of the formula, we predict the escape chances in the free falls of the three sets of unequal masses (Table II). The mean error in the five fractions is 1.7 to 1.8 percentage points, against 4.2 for the volume theory [2], 2.5 for the flux without absorptivity, and 1.0 to 1.6 for the flux with the absorptivity measured by simulation [4]. The largest deviation is in the heavy body of the set (3, 3, 1), which the formula overestimates, as the measured absorptivity already did.

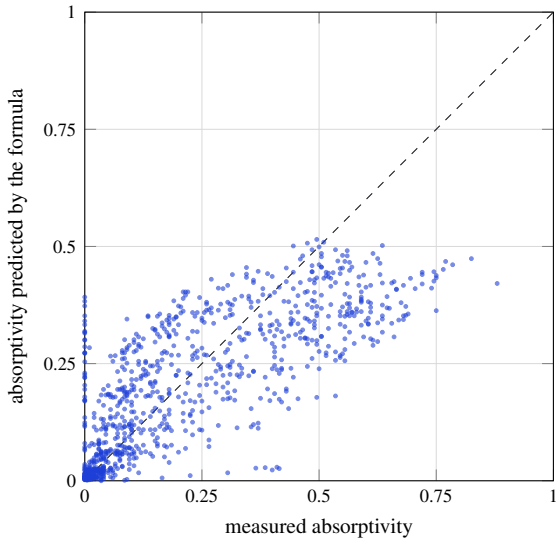


Fig. 1. Absorptivity predicted by the formula with eccentricity against the one measured by simulation, in 924 energy and eccentricity cells of seven mass channels that did not take part in the fit. The dashed line marks equality.

E. The final eccentricity remains open

The distribution of the final eccentricity is where the volume theory errs the most and where the measured absorptivity does best [4]. The formula falls in between (Table III). With equal masses, it takes the predicted mean eccentricity from 0.786, that of the pure flux, to 0.734, against 0.716 measured; the measured absorptivity gave 0.718 to 0.723. The shape of the distribution, however, still differs (Kolmogorov-Smirnov distance of 0.09, against 0.01 to 0.02 with the measured absorptivity). In the unequal sets, the formula does well for (1, 1, 3) and (1, 2, 3) and errs for (3, 3, 1).

IV. DISCUSSION

The piece of Kol's theory that depended on simulation can be approximated by an eight-constant formula, built from two simple physical rules: a mass threshold decides whether there is a partner exchange, and the energy of the pair, together with the change in the mass of the pair in the exchange, decides whether the expelled body stays bound. Used without any simulation of the tested cases, the formula predicts the escape chances with a mean error of about 1.7 percentage points, better than the flux without absorptivity and than the volume theory, and close to the result with the measured absorptivity. In practice, this means that the table of chances of who is expelled, for any masses in the regime studied, can be computed with an integral instead of hundreds of thousands of simulations.

What is missing is the eccentricity. Its effect on the absorptivity is strong in some channels, such as that of a light body facing a heavy, nearly circular pair, and almost null in others, and the speed of the pair at apoapsis captures only part of it.

Table I. Fraction of the variation among cells explained (R^2) by the laws of each part, predicting a mass combination left out of the fit, within the mass grid. Statistical noise limits the maximum to 0.93 and 0.94.

Part	Variables	R^2 without eccentricity	R^2 with eccentricity
Flyby or exchange	mass ratio	0.78	0.88
Staying bound, flyby	energy and masses	0.41 to 0.48	0.54
Staying bound, exchange	energy and D_0	0.72	0.73

Table II. Fraction of escapes by body in the free falls with five or more scrambles, compared with the predictions of the flux multiplied by the formula (without and with eccentricity), of the pure flux, of the volume theory in planar version, and of the flux multiplied by the measured absorptivity (notes FR-TC-09 and FR-TC-10).

Masses	Body	Measured	Formula	Formula with e	Pure flux	Volume	Measured absorptivity
1, 1, 3	mass 3	0.026	0.020	0.019	0.036	0.007	0.021 to 0.023
3, 3, 1	masses 3	0.055	0.086	0.086	0.100	0.022	0.072 to 0.074
1, 2, 3	mass 1	0.856	0.831	0.833	0.821	0.936	0.829 to 0.841
1, 2, 3	mass 2	0.125	0.131	0.130	0.128	0.053	0.139 to 0.151
1, 2, 3	mass 3	0.018	0.038	0.037	0.051	0.012	0.020

Table III. Mean eccentricity of the final pair in the free falls with five or more scrambles, measured and predicted by the flux multiplied by the formula with eccentricity, and Kolmogorov-Smirnov (KS) distance between the distributions.

Masses	Measured	Formula	KS
1, 1, 1	0.716	0.734	0.091
1, 1, 3	0.767	0.776	0.038
1, 2, 3	0.779	0.780	0.074
3, 3, 1	0.857	0.791	0.151

That is why the formula improves the mean eccentricity but does not reproduce the shape of the distribution. The structure of the arrival maps of note FR-TC-10 suggests the reason: with an eccentric pair, the outcome of the passage depends strongly on the orbital phase at which the body arrives, and an average over the phases hides regions with opposite behaviors [5].

The main limitations are as follows. All the laws are empirical and hold for the plane with zero total angular momentum and for the ranges of masses, energies, and eccentricities of the grid. The formula ignores direct chaos in the first passage and treats the choice of the expelled member as a coin toss, which is approximate when the incoming body is light. The final test uses three mass sets, and the deviation in the set (3, 3, 1) shows that the accuracy varies among channels.

V. CONCLUSIONS

The absorptivity of Kol's flux theory, in the plane with zero angular momentum, can be approximated by an eight-constant formula that combines a mass rule (flyby or exchange) and an energy rule (staying bound or escaping). Without simulating the tested cases, it reproduces the measured absorptivity with R^2 of 0.72 to 0.77, on average over eccentricity, and predicts the escape chances with a mean error of 1.7 percentage points. The dependence on eccentricity still has no general law, and the distribution of the final eccentricity remains the part of the prediction that requires the measured absorptivity.

TRANSPARENCY AND USE OF ARTIFICIAL INTELLIGENCE

The author is a lawyer and not a physicist. This work results from independent research, conducted with the assistance of artificial intelligence (a language model) in the literature searches, the programming of the simulations, the analysis of the results, the writing, and the translation. The author formulated the questions, followed the experiments, and is responsible for the published content.

DATA AND CODE AVAILABILITY

All numbers were obtained with scripts published at https://fisica-revisitada.vercel.app/en/code/, which run on Node.js without external dependencies and allow the results to be reproduced.

CONFLICT OF INTEREST

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[1] B. Kol, Flux-based statistical prediction of three-body outcomes, Celest. Mech. Dyn. Astron. 133, 17 (2021). doi.org/10.1007/s10569-021-10015-x

[2] N. C. Stone and N. W. C. Leigh, A statistical solution to the chaotic, non-hierarchical three-body problem, Nature 576, 406 (2019). doi.org/10.1038/s41586-019-1833-8

[3] V. Manwadkar, A. A. Trani, and B. Kol, Measurement of three-body chaotic absorptivity predicts chaotic outcome distribution, Celest. Mech. Dyn. Astron. 136, 4 (2024). doi.org/10.1007/s10569-023-10174-z

[4] W. W. M. da Cunha, Física Revisitada, Technical Note FR-TC-09 (2026). fisica-revisitada.vercel.app/artigos/memoria-e-acaso/

[5] W. W. M. da Cunha, Física Revisitada, Technical Note FR-TC-10 (2026). fisica-revisitada.vercel.app/artigos/evolucao-e-absortividade/

[6] W. W. M. da Cunha, Física Revisitada, Technical Note FR-TC-11 (2026). fisica-revisitada.vercel.app/artigos/regras-de-massas/